

# Vinod Kumar Ambala

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## Professional Summary

- 6+ years of experience building and deploying full stack and backend applications in enterprise environments.
  - Specialized in Java (Spring Boot), Node.js, Angular, and Python for scalable API and service development.
  - Designed and deployed cloud-native systems using AWS (EC2, Lambda, S3, ECS), Docker, and Kubernetes.
  - Integrated LLMs and GenAI tools (OpenAI, Hugging Face, LangChain) into production systems for NLP-based automation.
  - Automated CI/CD pipelines using Jenkins, GitHub Actions, and Docker to support fast, stable deployments.
  - Built secure RESTful APIs using JWT/OAuth2 and implemented microservices with resilience patterns, monitoring, and structured logging.
  - Delivered backend solutions across finance, logistics, and healthcare domains with clear performance and scalability gains.
  - Wrote clean, testable code with strong debugging practices, ensuring reliability in production systems.
  - Developed ETL pipelines, real-time data workflows, and ML-ready datasets using Python, AWS Glue.
  - Integrated GenAI features using OpenAI and Hugging Face APIs, enabling real-time NLP capabilities in customer-facing applications.
  - Led backend teams in architecture planning, code reviews, and release cycles for high-traffic applications.
  - Developed ETL pipelines and data services using Python, AWS Glue, and SQL to support analytics, reporting, and ML workloads.
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## Technical Skills

**Programming Languages:** Java (8–17), Node.js, Python, JavaScript, TypeScript, SQL,

**Frontend Development:** Angular (v8+), React.js, Vue.js, HTML5, CSS3, Bootstrap, jQuery, UI/UX Design

**Backend & API Development:** Spring Boot, Spring MVC, Spring AOP, Node.js (Express.js), Django, RESTful APIs, GraphQL, WebSockets, JWT/OAuth2, SOAP, .NET Core

**Microservices & Architecture:** Microservices, API Gateway, Circuit Breaker, Retry Logic, Load Balancing, Modular Monolith Refactoring

**Cloud Platforms & DevOps:** AWS (EC2, Lambda, S3, ECS, IAM), Azure, GCP, Docker, Docker Compose, Kubernetes, Terraform, CloudFormation, Jenkins, GitHub Actions, CircleCI, GitLab CI/CD

**Databases & Storage:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Snowflake, SQL Server, Oracle

**Monitoring, Security & Observability:** Prometheus, Grafana, ELK Stack, New Relic, OAuth2, API Security

**Data Engineering & Big Data:** ETL, Informatica, AWS Glue, PySpark, Data Structures, Snowflake, Kafka, RabbitMQ

**AI / GenAI & Machine Learning:** OpenAI API, Hugging Face Transformers, LangChain, LLM Integration, Prompt Engineering, FAISS, Pinecone, NLP, ML Model Pipelines, Scikit-learn

**Testing & Debugging:** JUnit, Mocha, Jest, TestNG, Selenium, BlazeMeter, JMeter, LoadRunner, Unit/Integration Testing, Performance Tuning, Debugging

**Development Tools & IDEs:** Spring Tool Suite, IntelliJ IDEA, Eclipse, VSCode, RStudio, Azure Data Studio, MySQL Workbench, MongoDB Compass, Postman, Tableau Desktop

**Collaboration & Practices:** Agile (Scrum), TDD, BDD, SDLC, Jira, Git, Bitbucket, Code Reviews, Version Control, Confluence

**Low-Code / No-Code :** Power BI, Power Automate, Tableau, ServiceNow, Salesforce, Zapier (exposure)

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## Professional Experience

### Software Developer

Dec 2024 - Present

Tigon Golf Carts

Philadelphia, PA

- Developed modular backend services for a Dealer Management System (DMS) using Spring Boot, covering customer, payroll, technician, and scheduling workflows.
- Designed and secured RESTful APIs with JWT authentication and RBAC; integrated SendGrid for automated communication workflows.
- Refactored legacy codebase into reusable microservices, improving maintainability and enabling faster feature delivery.
- Modeled and optimized MongoDB collections to support high-volume dealer operations, reducing query latency by 30%.
- Deployed cloud-native services on AWS (EC2, S3, IAM), improving system uptime and scalability.
- Collaborated with Flutter developers to define clean API contracts and ensure responsive cross-platform UI behavior.
- Enabled end-to-end automation of dealer operations, contributing to a 40% increase in customer onboarding and 60% drop in support escalations.
- Implemented CI/CD pipelines using GitHub Actions and Docker to automate builds, tests, and deployments across development and staging environments.
- Wrote unit and integration tests using JUnit and Postman to ensure API reliability and reduce regression issues during sprints.
- Participated in Agile ceremonies, provided backend estimates during sprint planning, and worked cross-functionally with QA, product, and design teams to align on deliverables.
- Developed a mobile-accessible web app with real-time notifications using **Firebase** and **WebSockets**, enabling instant dealer updates.
- Implemented background processing with **Node.js Worker Threads**, improving system throughput for async operations.
- Supported real-time data communication between backend and client, achieving seamless push notification delivery and enhanced UX.

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### Data Engineer

Jul 2024 - Nov 2024

KI Infosystems LLC

Saint Louis, MO

- Developed scalable ETL pipelines using **Python** and **AWS Glue**, automating ingestion and transformation of millions of records across multiple datasets.
- Integrated **AWS Lambda** for event-driven processing and scheduled workflows, reducing manual pipeline triggers by 100%.
- Managed raw and processed data storage using **Amazon S3**, ensuring cost-effective and secure data retention.
- Designed optimized **PostgreSQL** and **MySQL** queries to support analytics dashboards and downstream ML models.
- Implemented automated data validation scripts to monitor schema changes, missing values, and pipeline failures, improving data quality SLAs.
- Collaborated with analysts and data scientists to restructure raw data into ML-ready formats, accelerating experimentation cycles.
- Tuned Spark job configurations and partitioning logic (via Glue) to reduce job runtimes by up to 40% on high-volume datasets.
- Documented data lineage and transformation logic for business stakeholders, ensuring transparency and compliance readiness.

- Integrated upstream APIs and third-party sources using Python scripts to fetch and normalize external datasets for internal reporting use cases.
- Used AWS CloudWatch and custom logging within Glue jobs to proactively monitor ETL health and reduce recovery time from failures.
- Participated in sprint planning and backlog grooming sessions, contributing to cross-team alignment on data priorities and pipeline enhancements.

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## Software Engineer

Aug 2019 - Nov 2022

Savina Software Pvt. Ltd.

Hyderabad, India

- Developed core modules for HRMS and insurance applications using **Java 11**, **Spring Boot**, and microservices architecture.
- Built and maintained RESTful APIs for employee onboarding, payroll, leave management, and insurance claim processing.
- Integrated **Hibernate ORM** with **MySQL** to streamline data access, and optimized queries to improve transactional throughput.
- Wrote unit and integration tests using **JUnit** and **Mockito**, improving backend code coverage and reducing bugs in UAT.
- Automated monthly payroll reports using scheduled backend jobs, eliminating manual processing and reducing payroll cycle time by 60%.
- Collaborated with frontend developers working in **React.js** and **JSP**, ensuring seamless API integration and consistent UI behavior.
- Deployed services to staging and production environments; contributed to CI/CD practices using Git and Jenkins.
- Participated in daily standups, sprint reviews, and retrospectives, supporting Agile delivery and continuous improvement.

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## Software Engineer

Nov 2018 - July 2019

Lupin India

Maharashtra, India

- Participated in end-to-end SDLC phases including requirements gathering, development, testing, and deployment of internal systems.
- Assisted in backend development using **Core Java** and **.NET**, contributing to new features and system enhancements.
- Performed API and web services testing using **Postman**, validating REST/SOAP integrations for reliability and correctness.
- Conducted performance testing using **JMeter**, **LoadRunner**, **BlazeMeter**, and **NeoLoad** to ensure scalability and responsiveness.
- Collaborated with QA, developers, and business analysts to identify and resolve defects, supporting smoother release cycles.
- Documented technical test cases and debugging steps to assist future regression testing and system audits.

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## Education

Master's in Data Science

Jan 2023 - May 2024

Maryville University of Saint Louis

St. Louis, USA

Bachelor of Technology in CEE

Sept 2014 - May 2018

Jawaharlal Nehru Institute of Technology

Hyderabad, India

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## Key Achievements

- Built microservices with Spring Boot improving system maintainability by 30%.
  - Optimized Node.js backend response time by 25%.
  - Integrated Kafka for real-time streaming, reduced latency by 40%.
  - Developed REST APIs enhancing data flow and efficiency by 20%.
  - Improved PostgreSQL and MongoDB query performance by 25%.
  - Implemented CI/CD with Jenkins reducing deployment time by 40%.
  - Led integration of Node.js backend with Flutter frontend, enhancing UX by 15%.
  - Architected full-stack solutions from scratch for startups ensuring scalability and flexibility.
  - Wrote over **27,000 lines of production code** in 7 months, delivering scalable backend modules under aggressive deadlines.
  - **Introduced multithreading** using Node.js **Worker Threads**, enabling parallel processing and improving performance for compute-heavy tasks.
  - Refactored an entire legacy monolithic codebase into a **modular layered architecture**, improving testability, maintainability, and team productivity.
  - Architected and implemented a **clean, extensible project structure**, including folders for constants, utils, error handling, AWS integrations, authentication, environment configs, and background workers.
  - Developed a **web application consumed via a mobile wrapper**, enabling Chrome Extension-like functionality on mobile devices.
  - Built **real-time notification features** using **Firebase Cloud Messaging (FCM)** and **WebSockets**, enhancing user engagement and reducing latency.
  - Designed and delivered **custom backend services** aligned precisely with evolving product requirements.
  - Set up and maintained **robust CI/CD pipelines** with GitHub Actions and Docker, enabling automated deployments and quick rollbacks.
  - Created 78+ new backend files and modified 93+ existing ones as part of backend service development, refactoring, and modular architecture implementation.
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## University Projects

### Signal Detection for Adverse Drug Reaction Using Twitter Data

- Annotated 1M+ tweets for drug-event combinations.
- Achieved 96.6% confidence in prediction model.
- Tech: Spark NLP, Spark ML, MySQL, Python, Twitter API.

### Stock Movement Prediction Using Sentiment Analysis on Financial News

- Predicted intraday movements with 71.2% accuracy.
- Tech: Sentiment Analysis, Python, Web Scraping.

### Course Registration System

- Built a scalable system supporting 1000+ students.
- Tech: PL/SQL, DBMS.

### Bank Marketing Prediction

- Achieved 92% precision predicting term deposit subscribers.
  - Tech: Python, Decision Trees, SVM, Random Forest.
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## **Additional Information**

- Experienced in Java, Node.js, Python, Spring Boot, MongoDB.
- Participated in Agile practices (Sprint Planning, Standups, Retrospectives).
- Consistent learner; adapts to emerging technologies.
- Collaborates cross-functionally with teams across the stack.
- Strong problem-solving abilities with proven results in system optimization.
- Excellent communication skills for technical and non-technical discussions.
- Delivered systems with 20%+ performance gains and 30% faster processing.